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Clinic Management System

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CHAPTER 1: Introduction

Particularly in most medical clinic facilities, daily clinic transactions are still done on paper. We all know that modern clinics are now operating at great pace striving to serve as many patients as possible with the best of their abilities. But as the years rolled by, the number of patients has grown and various medical cases arise that the manual method of managing patients' records, prescriptions and appointment schedule, is no longer practical.

In this study, we hope to develop a web-based application that will minimize all paper works and manual records keeping, therefore allowing doctors and employee ease in keeping track of patients, reducing patients' waiting time and increasing the number of patients served – a system that is fully automated, user-friendly, time effective and efficient.

1.1 Background of the organization

Kombolcha Institute of technology(KIoT) is established in 1997 E.C.it gives treatment only for students when they get sick.it employee doctors,nurses,administrator,pharmacist and staffs.it gives treatment and medication for those students who learn in the institute.they have qualified doctors.the patient gets treatment when they go to clinic and also can make appointment if there need.

The clinic works 24 hours a day and also provides you the best medical care as possible,offer us with qualified and well trained doctors and physicians to serve us.the clinic gets its charge from the institute .

They do all the process to give treatment using manual system . the clinics are willing to work with us because they wants a technological system to make their work more simple and easy to manage the database.The clinic needs accurate clinic management system we asked them about the features included in the system.

1.2.Existing System of the project

Currently there is no clinic management system at work which uses computerized system . almost all clinics uses manual system for recording data about patients, staffs or doctors.it uses paper work and hand write method to record information , to make the bill record for payment ,to make appointment,to order medicine for doctors .all this made the patients wait a time maybe an hour or a day to get treatment as the way they did it takes more time and also treats few patients .

the proposed CMS works by changing all this methods of work to computerized way .as compared to the new CMS the existing system have several merits and demerits .if we see the merits of the existing system it is important if their is not enough internet access within the area as it does not stop their work since they execute manually.on the other hand it has several drawbacks as compared to the proposed system.this includes it uses a large space for storage of recorded data, takes a long time for recording data,difficult to back up stored data once it has lost .

1.3.Statement of the problem

Almost all or most of the clinics including KIoT student clinic use manual system which captures and manage data manually.If we see the analogue system it uses large space to store the recorded data.it is also difficult to back up and recover once data has lost.if some one loses his identification card one patient gain new id card it creates duplication of data. Patients also wait a time to get treatment because all those processes from registration to the final medication information is exchanged between doctors and staffs using paper works.

Generally usage of large space for storing data,redundancy of data,time consuming of manual system ,costs it uses to cover for paper works,back up of data if lost are the problems of clinics.

Problem of storage comes from storing a large documents of recorded data that are done manually using paper works .so if there are many paper works it uses large space .

problem of redundancy of data arises from registration of patient that lost or forgot password and they come to clinic many times.

problem times comes from registration to final treatment and medicine it takes a long time .
Problem of cost arises from money lost to buy papers it cannot make them more profitable .
Problem of back up :it comes from conditions if data is lost and how can we recover those lost data.

1.4.Proposed System

CMS system important to alleviate the above stated problems .this system includes patient registration form for new patients and use ID number for login into the system ,log in page works for employees and patient .it can register data store data and we can retrieve it if we need it later .so registration, order of medicines to staffs from doctors are made using computer .it needs internet access.

1.5 General Objective and specific objective

The general objectives of our project is to develop CMS for KIOT students clinic.

Specific Objectives

To do the general objectives of CMS we do the specific tasks:

- To gather requirement
- To requirement specification
- To requirement modeling
- To Implement CMS
- To design and develop a CMS for KIOT clinic
- To carry out system problem for possibility of developing CMS for improving the system

1.6 scope and limitation of CMS

1.6.1 scope

We do the system for KIoT students clinic, this system solve problems under this area .the system can work for other clinics by increasing additional features and modules.This system has the following system :

Patient appointment :in this module can add,edit appointments that make by patients.the employee can only to add new appointments in black slot.the doctor then check back the appointment that was made by him/herself by using dates to filter the appointment.

Patient information:in this module employee can add and edit patient information which consists basic information about patient.the doctors can update patient information.

Report:this module is prepared for patient medication report , patient dispensary report which includes list of medicines and prescription .the employee then gives medicine for patient based on this dispensary report.in this module we will record all clinic employee member information.once the employee has registered in the system,he/she is allowed to login to the system. Employee information can be edit and update by admin.

1.6.2 Limitations

Our system has the following as some limitations, referral, emergency,inventory,laboratory and reception module.

1.7 Methodology for the project

We use the following requirement gathering methodology :

Observation:before we design a system to the clinic,we do visit the clinic to do an observation how they operate to treat a patient.it is an ordinary clinic which does its work manually with out help of technology.we observe and record in detail how a progress goes from patient registration until the patient gets medicines from dispensary counter.We see that a lot of paper is needed a lot of paper to do the progress.

Interview: end users of the system are those who have interacted with the system, so we can get information about requirements. They tell us what are the requirements to do the system and tell us clearly what they expected from the system. So interviewing the end users of the system gives us clear and definite requirements of what they want from the system.

Questionnaire: we prepare questions for clinic employees and they answered the questions.

1.8 Feasibility Study

A feasibility study is an analysis that considers all of a project's relevant factors including economic, technical, legal, and scheduling considerations to ascertain the likelihood of completing the project successfully.

Operational feasibility: The new system project will permit the clinic administration system to reduce time absorbed and to bring the arrangement in short time and delivering productivity results in a powerful way. Online registration and recording data in a database enables the system to be more effective in bringing out the best result. The new system project permits the clinic management system to give accurate data and store them in a database which will hold the data in a secured path without losing any data. So it is operationally feasible.

Economic feasibility: It is economically feasible because it does not require much money. It needs a computer for documentation and implementation and testing of the system. So as it doesn't cost much, we did this system with little money.

legally feasible : The CMS is also legally feasible because it does not conflict with the laws of the country. It does not do anything which contradicts the existing rules and regulations of the government.

Technical Feasibility: Since it is not a complex system, we have the technical feasibility of developing the system.

1.8.1 Significance of the project.

Our project has the following contributions to clinics, patients and clinic employees.

- The Enable better supervision of employees.

- Reduce operational cost.since a single system will be set to monitor and process loads of the institution's operation, this will cut down on the unnecessary spending. The rest of the capital will help in improving the institution facilities and other amenities.
- Reducing errors. Our software will provide proper validation techniques that will collectively reduce errors done when recording data.
- Increased efficiency and effectiveness of work for employee.
- Enabling ease of access to information concerning each patient and their previous records thus guaranteeing efficiency and effectiveness.
- get treatment with given time interval for patient.

Chapter 2: System Requirement Specification

2.1. Background (Overview)

A requirement specification is a collection of all requirements that are to be imposed on the design and verification of the system. The specification also contains other related information necessary for the design, verification, and maintenance of the system. Requirements specification collects and formalizes the information about the application domain and expected functions.

The input is the set of business requirements that motivate the application development and all the available information on the technical, organizational, and managerial context. SRS of CMS describes about functional and non functional requirements how it works and what the system do and the purpose of requirements. we collect and formalize about application domain and expected functions . the clinic motivates system development gives available information and it helps us as input. the SRS documents including scope, purpose, interface, analysis model etc.

2.1.1 Scope

The CMS includes the following modules. It contains a module for registration for new and already registered patients and employees can see the patient information ,also a module for set appointment for patients and a doctor can see these appointment. It also includes a module for patient information to add new patient ,edit ,update patient information.

2.1.2 Purpose

The CMS is intended to provide a system that provides solution through system using internet. we aim to develop a system that meets the requirements stated and needed by the user with the required quality for patients that overcomes the drawbacks of the manual system to get good treatment within the right time. once we developed this CMS with required quality users have used it to get services easily. we are intended to develop which is easily understandable by users. the purpose of our system is to provide good system that matches the users requirements with right qualities such as clarity, accuracy, completeness, and consistency of the system to be developed.

It also checks verifiable At the end of the project, the user/client will be able to verify that all that all the agreed deliverable have in fact been produced and meet the project

management requirements specified. Modifiable is one and the SRS can be modified when the development team and user feel the need.

2.1.3 Intended Audience and Suggested Readings

The intended audience of this document would be the students and specific employees like advisor, consultants and System Operators of kiOT, and project team, supervisor with the objective to refer and analyze the information. The SRS document can be used in any case regarding the requirements of the project and the solutions that have been taken.

The document would finally provide a clear idea about the system that is building. Our document is intended for reading to users, employee, patients, testers, admin and doctors, and they give us what we have missed or not include requirement, unnecessary addition and definition of stated requirements they give as feedback to make expected system.

2.1.4 List of Acronyms, Abbreviations and Definitions

Abbreviations	Full form
CMS	Clinic management system
ID number	Identification number
SRS	Software requirement specification
KIOT	Kombolcha institute of technology
OS	Operating system
UI	User interface

Table 2.1 list of abbreviations

Word/ term	definitions
Employee	Workers of clinic such as admin and doctor
Front end	Those language used to develop user interface page for the users to communicate with system like html, css, js

Table 2.2 list of definition of words

2.2. Overall Description of Software Requirements

2.2.1. Product Perspectives

Product perspective is essentially the relationship of the product to the other products, defining if the product is independent or is part of a larger product (dependent), and what the principal interfaces of the product are. This system is totally independent system that manages activities of the CMS as taking appointments, generating patient reports. In this project all the records are stored in single database. Different users have different permission to access this web application. Each user has unique ID. If any data is lost user is having option to recovery. User's don't have right to alter records after particular time period and also it is not having option to alter other patient records.

The CMS is the replacement of the existing manual system of the clinic.it works by replacing paper works for all progresses of patient treatment,all activities of paper works are in computerized way.

2.2.2. User characteristics

The admin, doctors and patients will be the main users. The system is also designed to be user-friendly.

- Admin
- Employee
- Doctors
- Patients

Admin: Admin should have prior knowledge of the system. Admin is able to control the whole system. He/she can add , update and modify the system.

Doctor: should fairly know about the usage of the system. Doctors are able to see the respective appointments taken. And also can view patient's details and records. And also can take appointment online.

Patients: also use the system for registration to see their detail information.

2.2.3 User Documentations

The CMS has user manual which describes about the system .it describes about how the system looks like ,how to use the system ,contains description included by giving the document copy give proposal ,for users,sending it via email.the manual consists product description,technical detail and how to use it and contact via email.

2.2.4 Assumptions and Dependencies

The following are assumptions :

- the system needs correct username and password to login into the system from the user .
- The system must be user friendly for users to use.
- The user must know about the system to use and be aware of computer.
- The system needs computer and connection for internet maybe transmitted via cable or direct WiFi.
- Data must be inserted correctly for update,edit data within the system.
- The system is available 24 hrs a day for users if there is internet connection.
- The codes must be correct or error free to function correctly.
- The user entered into the system within 10 seconds when he login in the system.
- Information must be stored in database and must retrieved when needed fast and quickly once connected to the system.

The following are dependencies :

- The system depends on computers, internet connection and windows operating system.
- The system depends in the awareness of the patient, employee and doctors.
- The system must be developed based on the users requirement and needs and required qualities.

2.2.5 General constraints

Software constraints: computer speed and and operating system.

Hardware constraints: lack of hardware materials,internet access are some constraints of CMS .

2.3 System Requirements

System requirements are the configuration that a system must have in order for a hardware or software application to run smoothly and efficiently. Failure to meet these requirements can result in installation problems or performance problems.it's a detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented.

2.3.1 Functional Requirements

Functional requirements capture the intended behavior of the system. This behavior may be expressed as services, tasks or functions the system is required to perform.Functional requirements describe what the system or software must do and sometimes called behavioral or operational requirements.

What the system is describes to provide, How the system should respond or react to particular input or situation,What the system should not do,Constraints on implementing the above said requirements

Functional requirements of the the CMS are

Registration Process

Adding Patients: The clinic Management must enables the employee in the front desk to include new patients to the system.

Assigning an ID to the patients: The CMS should enable the employee in the front desk to provide a unique ID for each patient and then add them to the record sheet of the patient. The patients can utilize the ID throughout their clinic.

Report Generation

Result and prescription report: CMS must generate a report on every patient regarding various information like lab report, medicine and prescription.

Patient information

Patient Information: Every patient must have some necessary data like phone number, their full name, country, address, city, 'patient's ID number, etc.

Updating information of the Patient: The clinic management system must enable doctors to update the information of the patient as described in the mandatory information included.

Patient appointment

CMS should enable employee to make new appointments and also the doctor sees this appointment. The system must also allow to update these appointments.

Employee information

CMS should enable admin to add new employee and update employee information.

2.3.2 Non Functional Requirements

Usability of the requirement

The system should be very straightforward to the user and easy to operate by the user. The error message that is shown when an error is detected must be in an understandable language to the user. Easy to use because, the system information only displays in English language for easy understanding.

Efficiency requirement

The system must allow the employee to search the patient record in an easy and efficient way from a large amount of data. The system response time must be fast and the system should allow the user to open several tasks at one time when they are using it.

Reliability requirement

The error rate of this system must be shrinking. adequacy error message should prompt to the user when there is any validation error occurred.

Availability

The system is available all the time 24/7.

Security requirement

The system must provide highly security on protecting patient privacy. some confidential data should restrict to only authorize user to access it by using username and password.

Maintainability

Back-Up: The system offers the efficiency for data back up by storing hard disk and secondary memory.

Errors: The system will track every mistake as well as keep a log of it.

the system should not fail more than one in a month.

Safety

- In case the user forgets or loses Password, the repair functionality helps by choosing “forgot password” option in the main login window.
- To avoid this kind of situations, backups can be done regularly.
- While typing the password, if the caps lock is on it must be notified.
- If the system is kept idle for 10 min the session will expire. During the maintenance stage. SRS document can be referred for any validations.

Portability: This system can be used in any personal computers supporting windows operating system platform.

Flexibility: The system keeps on updating the data according to the transactions that takes place.

Timeless: The system carries out all the operations with consumption of very less time.

Security:

Performance: the system shall responds to the user within less than 10 seconds.

2.3.3. Hardware and software requirements

System Design of Clinic Management Service:

Now this clinic management system is designed in such a way that it takes fewer resources to work properly.

It has its own sort of minimum requirements that we need to take care of:

- The system needs a minimum of 2 GB of ram to run all the features smoothly and suddenly.
- It needs a minimum 1.3 GHz processor to run smooth as less than that may create problems.
- The system needs to be operated by some authorized person as wrong hands can make it irresponsible.
- Rest is all up to the user's usage will care for hardware.
- For security Antivirus is recommended.

The system is made properly and all the testing is done as per the requirements. So, the rest of the things depend upon the user and no one can harm the data or the software if the proper care is done. All the attributes are working perfectly and if any error is found then it can be removed easily.

It is all about a document or set of documentation that describes the features and behavior of a system or software application

2.4 External Interface requirements

All the interactions of the software with patients, doctors, hardware and software are specified here.

2.4.1. User Interfaces

The user interface is designed in front end . The developer will have to study the designing of the product. The use of the controls and the component from the Add items feature of the front end. The user of the product will get very user friendly web page which will be very easy to work with.

2.4.2. Hardware Interfaces

The one used here is monitor, keyboard and mouse.

The system should have these hardware:

- Processor
- Memory
- Hard Disk Drive

2.4.3. Software Interfaces

- Operating System: Windows OS

2.4.4. Communication Interfaces

Communication is done through internet and intranet and communicate using browser to communicate with the system .once the user entered username and password the browser holds this data make a change in a form understandable to the database ,and when it gets its username and password and validates it ,it allows a user to log in to the system and do their task once they finish their work they can logout from the system/site .so communicate using browsers like chrome ,fire fox or opera and other browsing things.

2.5 Analysis Models

2.5.1 use case diagram

Use Case Diagram for Clinic management system: The main actors of clinic management system are admin, system users,doctor,patient ,who perform different type of use cases such as manage clinic,manage patient ,manage appointment,manage patient profile ,manage report.

The relationship between use case actors of clinic management system:

Patient entity:use cases of patients are request for registration ,view appointment,view report.

Doctor entity:use cases doctor are view appointment, generate report,make appointment, add new patient and update patient information.

Admin entity:use cases of admin are mange account ,update employee information and add new employee.

Figure 2.1 use case diagram for CMS

Use Case scenario description for CMS

Use Case Name: View appointment

Brief Description: This use case allows user to view appointment information.

Actor: Patient and Doctor

Precondition: the user must be login.

Post condition: the user must logout.

Main Flow:

1. the user login into the system
2. Verify password and ID
3. Click the appointment page
4. Display appointment

5. Logout

Alternate flow:

1. invalid username and password and prompt error message.

Use Case Name: report generate

Brief Description: This use case allows doctor to generate report.

Actor: Doctor

Precondition: the doctor must be login.

Post condition: the doctor must logout.

Main Flow:

1. the doctor login into the system
2. Verify password and ID
3. Click the report generate page
4. Display report generate page
5. write report for result, prescription and medicine
6. click save button
7. verify report data
8. successfully saved
9. logout

Alternate flow:

2.invalid username and password and prompt error message.

7.invalid data and prompt error message

Use Case Name: update employee information

Brief Description: This use case allows admin to employee information.

Actor: Admin

Precondition:the admin must be login.

Post condition: the admin must logout.

Main Flow:

1.the admin login into the system

2.Verify password and ID

3.Click the update employee information page

4.Display employee information page

5.update employee information

6.click save button

7.verify updated employee information

8.successfully saved

9. logout

Alternate flow:

2. invalid username and password and prompt error message.

7.invalid information and prompt error message

Chapter 3: Requirement Analysis and Modeling

3.1 introduction

Requirements analysis and modelling is probably the most important skill for a business analyst. The success of any software project depends on this task. It includes modeling diagrams such as activity diagram, use case diagram, sequence diagram and class diagram.

3.2 Activity Diagram of clinic management system

The activity diagram of clinic management system which shows the flows between the activity of admin, doctors, patient,. The main activity involved in this activity diagram of clinic management system are as follow:

- Make appointment activity
- Update patient information activity
- Update employee information activity
- Add new patient activity
- View appointment activity
- Report generate activity
- Manage account activity
- Add new employee activity
- Create account activity

Features of the activity diagram of CMS

- Doctors, admin can create account.
- Doctors can view selected appointment and make appointments.
- Doctors can be able to generate report of patients.
- All objects such as Appointments and Doctors are interlinked.
- Patients can view appointment, create account and view report.
- Admin can add new employees, manage account and update employee information.

To do all this admin, doctors and patient must login to the system and finally logout from the system except for create account.

Figure 3.1 activity diagram for making appointment

Figure 3.2 activity diagram for view report

3.3 Sequence diagram for CMS

The sequence diagram of clinic management system which depicts the interactions between the objects of Doctor, admin and Patient. The following are examples of class objects used in this Sequence Diagram of a clinic System:

- Doctor Object
- Patient Object
- Admin Object

Login sequence diagram of CMS, where admin will be able to login in their account using their credentials.

Figure 3.3 sequence diagram for add employee information

Figure 3.4 sequence diagram for add new patient

3.4 Class diagram of CMS

A class diagram is a static model that shows the classes and the relationships among classes that remain constant in the system over time. The class diagram depicts classes, which include both behaviors and states, with the relationships between the class. clinic management system class diagram describes the structure of of a clinic management system classes, their attributes, operations (or methods), and the relationships among objects. the main classes included in this document are:

- patient class: manage all operations of patient.
- doctor class: manage all operations of doctors.
- Admin class: manage all operations of clinic.

Figure 3.5 class diagram for CMS

Conclusion

Finally, in the Clinic management system, the outcome of all the-the hard work done for the clinic management system is here. It is system that helps the user to work with the clinics easily. This system reduces the amount of manual data entry and gives greater efficiency. The User Interface of it is very friendly and can be easily used by anyone.

It also decreases the amount of time taken to write patient details and other modules. In the end, we can say that this system is performing the tasks and is doing the work for which it is made.

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